One computational system or two? Evidence from semantically incongruent propositions

Seyyed Hatam Tamimi Sa'd, PhD (shtamimi90@gmail.com)
Harry S Truman College, Chicago, IL

39th Annual Conference on Human Sentence Processing

Summary

Background
Bimodal bilinguals, fluent in both a spoken and a signed language, can produce simultaneous speech and sign, occasionally with mismatching word orders.

Objective: This study investigated whether such bilinguals could generate semantically incongruent sentences using Sadat Tawaher Sign Language (STSL) and Khuzestani Arabic (KhA).

Example: **Speech** I am not hungry. **Sign** I am full.

Results: Data from 13 fluent bimodal bilinguals (aged 7–56) produced: **(i)** frequent code-blending (95%), **(ii)** word-order mismatches (33%) **(iii)** no semantically incongruent sentences.

When prompted to produce conflicting propositions, participants failed, reporting high cognitive difficulty.

Discussion: Bimodal bilinguals likely operate with a single computational system:
- capable of differing surface word orders
- incapable of two semantically distinct propositions

Conclusion: Despite having two output modalities, bimodal bilinguals operate within a single computational system that allows multiple word orders but cannot encode two semantically distinct propositions.

→ likely due to cognitive & memory constraints

Language, Modality & Bilingualism

Language is typically produced in one of two modalities:

- auditory-aural modality → spoken
- visual-gestural modality → signed

Bilinguals are individuals competent in 2 languages.

- If individuals are competent in two spoken languages or two sign languages, they are **unimodal bilinguals**.

Unimodal bilinguals are restricted physiologically (setting some exceptions aside) because they can only produce one word / one sign at a time.

But … what if a person speaks and signs?

See next section →

Bimodal Bilingualism

Bilingual users of one spoken language and one sign language are **bimodal bilinguals**.

Two robust findings about bimodal bilinguals:
(i) strongly prefer **code-blending** (simultaneous use of speech & sign)

(ii) commonly avoid **code-switching** (switching from speech to sign & vice versa)

Finding & Debate

Finding: Bimodal bilinguals occasionally produce syntactically incongruent, that is, mismatching word orders

Italian:	Chi	ha	chiamato?	[Wh + Verb]
	who	have	call	
Italian Sign Language:	CALL		WHO	[Verb + Wh]
	'Who has called?'			

Donati & Branchini (2013: 118-119)

Debate: This finding has sparked a debate in the psycholinguistics literature over whether bimodal bilinguals have:

(i) a single computation / derivation that generates two word orders,

or

(i) separate, simultaneous computations or derivations, one per word order.

Current Study

Bimodal bilinguals have two output channels: **hands** & **vocal tract.**

Objective: I investigated whether bimodal bilinguals can produce semantically incongruent sentences, one via their hands (in Sadat Tawaher Sign Language) and one via their vocal tract (in Arabic).

Methods:
- **Languages**: Arabic & Sadat Tawaher Sign Language (STSL)
- **Participants**: 13 hearing Arabic-STSL fluent bimodal bilinguals (9 males, 4 females), aged 7-56 years
- **Data collection methods**: sentence production task, free narratives, & online sentence production task
- **Data**:
  - Step 1: 1,100 sentences collected off-line
  - Step 2: 20 semantically incongruent sentences collected live

Examples of semantically incongruent sentences:

	Speech	Sign
Pair 1:	I am not hungry.	I am full.
Pair 2:	My friend didn't buy a car.	My friend saved his money.

Sadat Tawaher Sign Language

STSL emerged around 60 years ago in the village of Sadat Tawaher in southwestern Iran after a 20-year-old man, named *Hanash*, lost his hearing. Sadat Tawaher is a small village with a population of 450 people. Arabic is the only spoken language in Sadat Tawaher. Hanash was / is the only deaf person in Sadat Tawaher. Hanash's family and friends relied on gesture to communicate with him due to lack of access to deaf education or exposure to sign language.

Thus far, 3 generations have leaned STSL as their L1, in addition to Arabic. Almost all signers speak when they sign. These signers therefore qualify as bimodal bilinguals.

STSL has a basic SVO order. Negation is typically sentence-final and wh-sign remains in situ.

PEOPLE HARVEST RICE
'The people harvested the rice.'

BOY PUSH WHO
'Who did the boy push?'

TEDx Talk on STSL

Findings

(i) Findings from Step 1:

- Patterns observed in sentence productions & narrative data:

a. **Frequent code-blending**: In 98% of utterances, signers used speech and sign simultaneously. 66% of utterances had matching orders in speech and sign:

Speech: SOLDIER SHOOT MAN

Sign: SOLDIER SHOOT MAN
'The soldier shot the man.'

b. Mismatching word orders: In 33% of utterances, signers used two different word orders in speech and sign:

Speech: BOY WHAT GAVE TO-GIRL

Sign: BOY GIVE GIRL WHAT
'What did the boy give the girl?'

(ii) Findings from Step 2:

- Patterns observed throughout live data collection sessions:

c. Even after several attempts, all participants failed to produce semantically incongruent sentences!

d. Instead, participants either (i) code-blended or (ii) code-switched.

See example on the right →

Example: Produce the following sentences simultaneously in speech & sign:

Speech: My friend didn't buy a car.
Sign: My friend saved his money.

Participants produced them either as code-blending:

(i) **Speech**: FRIEND NOT BUY CAR, FRIEND SAVE MONEY
Sign: FRIEND NOT BUY CAR, FRIEND SAVE MONEY

or as code-switching:

(ii) **Speech**: FRIEND NOT BUY CAR, switch to sign FRIEND SAVE MONEY

But never as:

(iii) **Speech**: FRIEND NOT BUY CAR
Sign: FRIEND SAVE MONEY

Discussion & Conclusion

None of the participants were able to successfully produce semantically incongruent sentences in speech and sign. They reported to have found the task extremely challenging.

In conclusion, bimodal bilinguals appear to have a single computational system despite having two output channels-vocal tract and hands.

This system sometimes generates utterances in two different surface word orders, but it almost never two semantically incongruent propositions.

Semantically non-equivalent propositions impose the greatest cognitive cost on the production system.

References

[1] Branchini, C., & Donati, C. (2016). Assessing lexicalism through bimodal eyes. *Glossa*, *1*(1), 48, 1-30. **[2]** Donati, C., & Branchini, C. (2013). Challenging linearization: Simultaneous mixing in the production of bimodal bilinguals. In T. Biberauer & I. Roberts (eds.), *Challenges to linearization* (pp. 93-128). Berlin: Mouton de Gruyter. **[3]** Emmorey, K., Borinstein, H. B., Thompson, R., & Gollan, T. H. (2008). Bimodal bilingualism. *Bilingualism*, *11*(1), 43-61. **[4]** Emmorey, K., Li, C., Petrich, J., & Gollan, T. H. (2020). Turning languages on and off: Switching into and out of code-blends reveals the nature of bilingual language control. *Journal of Experimental Psychology*, *46*(3), 443-454. **[5]** Emmorey, K., & McCullough, S. (2009). The bimodal bilingual brain: Effects of sign language experience. *Brain & Language*, *109*, 124-132. **[6]** Lillo-Martin, D., D. Gagne, M. Avino, J. D. Bobaljik, S. Wurmbrand, R. M. de Quadros, & G. Keller. 2025. Linguistic characteristics of bimodal bilingual code-blending: Evidence from acceptability judgments. *Bilingualism: Language and Cognition*, 1-14. **[7]** Quadros, R. M., D. Lillo-Martin, & D. Chen Pichler. 2020. Blending languages: Bimodal bilinguals and language synthesis. In *Sign Multilingualism*, ed. by U. Zeshan & J. Webster, 201-226. Berlin/Boston: De Gruyter Mouton.

Contact

Hatam Tamimi, PhD

shtamimi90@gmail.com